

Alyssa Langmeyer

Professor Corizzo

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## Media Bias Detection in Political News

The central question driving this project is whether a machine learning model can detect media bias purely from the language of an article, without any knowledge of the outlet that published it. A model that learns to associate a brand name with a bias label is not detecting bias; it is memorizing outlet identities. A genuinely useful classifier must learn what biased writing looks like at the linguistic level, independent of source. This project builds such a pipeline using the NewsMediaBias-Plus dataset, trains a text classification model on TF-IDF features, and then examines which specific words and phrases drive the predictions.

### **Dataset and Label Methodology**

The NewsMediaBias-Plus dataset (Raza et al., 2024), developed by the Vector Institute for Artificial Intelligence, contains 40,945 news articles drawn from 91 outlets ranging from major wire services such as AP and Reuters to partisan commentary publications on both ends of the political spectrum. The dataset is multimodal, pairing each article with its top image, though this project used only the text.

The target variable is the `nlp_label` column, a binary annotation indicating likelihood of disinformation: "Likely to be Bias" (coded 1) or "Likely to be Unbiased" (coded 0). These labels were generated through multiple LLM-based annotation strategies with content expert review, as part of the ViLBias framework (Raza et al., 2024). The annotation targets linguistic cues in the text itself: evaluative language, charged framing, selective emphasis, and the presence or absence of neutral attribution patterns.

It is important to clarify what "biased" means in this context. The label does not indicate factual inaccuracy. An article can be entirely accurate and still receive a bias label if its writing style matches the register of disinformation: strong adjectives, emotionally loaded framing, political labels used evaluatively. Conversely, neutral wire-service prose receives the unbiased label because its pattern of heavy attribution and plain verbs does not match the linguistic profile of disinformation. The dataset documentation includes a disclaimer that these classifications are intended for research purposes and should not be treated as definitive judgments about any article or outlet.

The dataset carries a 69/31 class imbalance, with biased articles representing the majority. A naive classifier could reach 69 percent accuracy by always predicting biased, without learning anything meaningful. Every modeling decision in this project was designed to prevent that outcome.

## **Data Cleaning**

Raw scraped news articles contain far more than article text. A typical scrape captures cookie consent banners, paywall notices, newsletter prompts, timestamps, author bios, and footer boilerplate alongside the actual content. All of that material is irrelevant to bias detection and was systematically removed.

The cleaning pipeline proceeded in six stages. First, the headline and article body were concatenated into a single text field. Headlines carry significant editorial signals and are frequently where slant is most apparent; treating them separately would discard that information. Second, URLs, HTML entities, and non-alphabetic characters were stripped via regular expressions. Third, all text was lowercased so that variant capitalizations of the same word are treated identically.

Fourth, a two-pass artifact removal process cleaned out scraping junk: subscription prompts, timestamps, cookie strings, and author attribution footers. Fifth, every mention of the publishing outlet's name was removed from the article text. This step is the most consequential decision in the project. If outlet names remain in the text, the model learns to associate brand identity with bias labels rather than learning anything about writing style. Removing outlet names forces the classifier to detect bias from linguistic patterns alone. Sixth, articles with fewer than 50 characters of remaining text were dropped as failed scrapes.

### **Feature Extraction and Model Training**

Text was converted to numeric vectors using TF-IDF (Term Frequency-Inverse Document Frequency). For each word or phrase, TF-IDF assigns a score reflecting how frequently it

appears in a given article relative to how commonly it appears across the full corpus. Distinctive vocabulary receives high scores; ubiquitous function words receive scores near zero. The resulting matrix contains 10,000 features per article.

Both unigrams and bigrams were included (`ngram_range = (1,2)`), capturing two-word phrases alongside individual tokens. This matters because many bias signals are phrase-level. "Far" and "right" carry little predictive weight individually; "far right" as a unit is a specific evaluative label that appears heavily in opinionated coverage and rarely in wire-service prose. Sublinear TF scaling (log normalization) prevents long articles from dominating the feature space, and `min_df = 5` filters out terms appearing in fewer than five articles.

A Logistic Regression classifier was trained on a 70/30 stratified train-test split, preserving the 69/31 class ratio in both partitions. The `class_weight='balanced'` parameter adjusts the loss function so that errors on the minority (unbiased) class are penalized proportionally more heavily, preventing the model from defaulting to the majority class prediction. Supplementary analyses included chi-squared feature selection, bigram-level vocabulary comparison, outlet-level bias averaging across all 91 outlets, and K-Means clustering ( $k=3$ ) to test whether bias emerges as a natural grouping dimension without supervision.

## Results

The model achieved an ROC-AUC of 0.82, meaning it correctly ranks a randomly selected biased article above a randomly selected unbiased one 82 percent of the time (chance = 0.50). F1 for the biased class was 0.80; F1 for the unbiased class was 0.64. The lower unbiased F1 reflects

the class imbalance directly: the model has seen fewer examples of neutral writing and makes proportionally more errors on that class. This is expected behavior given the data composition, which is why AUC serves as the primary evaluation metric rather than accuracy or per-class F1.

Feature coefficient analysis reveals the linguistic patterns driving predictions. Words most predictive of the biased class include Trump, Biden, Harris, White House, Supreme Court, far right, assassination attempt, and radical. These terms appear most densely in opinionated, politically charged coverage. Their presence does not inherently indicate bias; rather, the articles in which they appear are more likely to be written in an evaluative, advocacy-driven register. The model learned this correlation from over 40,000 labeled examples.

Words most predictive of the unbiased class include said, police said, according to, year old, gold medal, and paris olympics. These reflect the conventions of neutral journalism. "Said" is the AP Style Guide's prescribed attribution verb precisely because it carries no evaluative loading, unlike alternatives such as "claimed" or "insisted." Non-political topics like sports results and local crime reporting are consistently covered in a factual, attribution-heavy register, which the model learned to associate with the unbiased label.

### **Outlet-Level Findings and Clustering**

Averaging predicted bias probability across all articles from each outlet produces a ranking of 91 outlets by linguistic profile. The most striking result is that the highest-scoring outlets span the full political spectrum. The Federalist and Daily Caller (right-leaning) appear alongside ProPublica, Daily Kos, and The Guardian US (left-leaning). What they share is not

political orientation but writing style: all are opinionated, advocacy-driven, and use charged evaluative language. The least biased outlets are AP, Reuters, CNBC, and Bloomberg, whose writing culture prioritizes neutral attribution and factual precision. This confirms that the model is measuring opinionated versus neutral writing style rather than political direction.

K-Means clustering produced a V-measure near zero, indicating that articles do not naturally cluster along the bias dimension without supervision. TF-IDF vectors organize articles by topic first: Olympics articles cluster together regardless of their bias labels, as do Supreme Court articles. The topic signal overwhelms the stylistic signal entirely. This result demonstrates that labeled training data is necessary for bias detection; unsupervised methods cannot surface this dimension from raw text alone.

### **Limitations and Future Directions**

The model is only as reliable as its labels. LLM-based annotation, while reviewed by content experts, can carry biases from the underlying models used to generate it. Additionally, a word-frequency approach cannot detect bias through selective omission or misleading framing, both of which can occur in otherwise neutral-sounding prose. The class imbalance limits performance on the unbiased class, and a more balanced dataset would likely improve that F1 score.

Future work could fine-tune DistilBERT end-to-end on this task, which would capture semantic context beyond word frequency and likely push AUC into the high 0.80s or low 0.90s. A topic-stratified analysis could test whether bias patterns differ across coverage domains such

as immigration, economic reporting, and sports. The dataset also supports multi-class bias classification across political, gender, and racial bias dimensions, which would produce more specific outputs than the binary label.

## Works Cited

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